

Stiefel and Grassmann Manifolds: A Geometric Foundation for Orthogonality and Subspace Problems in Machine Learning

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Abstract

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1 Introduction

2 Preliminaries

We follow the conventions of [2] and [1].

2.1 Basic Definitions

Definition 2.1. Let X be a topological space. A **Co-ordinate Chart** on X is the pair (U, φ) where $U \subset X$ is open, and φ is a homeomorphism

$$\varphi : U \xrightarrow{\sim} \varphi(U)$$

where $\varphi(U)$ is an open subset $\varphi(U) \subset \mathbb{R}^n$, for some nonnegative integer n .

Definition 2.2. Let X be a topological space, and fix a nonnegative integer n . X is an **n-dimensional topological manifold** if and only if X is Hausdorff and second countable, and for any point $x \in X$, there exists a co-ordinate chart,

$$\varphi : U \xrightarrow{\sim} \varphi(U) \subset \mathbb{R}^n$$

where $x \in U$, and $\varphi(U)$ is open in $\mathbb{R}^n$.

This is to say, that X is locally Euclidean, i.e. every point of X has an open neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$.

We require that X is Hausdorff, and second countable, as to exclude certain pathological examples.

Definition 2.3. Let X be a topological manifold, and let (U, φ_1) and (V, φ_2) be two co-ordinate charts on X . The **transition function** between these two co-ordinate charts is the function:

$$\phi_{21} = \varphi_2 \circ \varphi_1^{-1} : \varphi_1(U \cap V) \xrightarrow{\sim} \varphi_2(U \cap V)$$

Notice, it is automatically a homeomorphism, as it is the composition of two homeomorphisms.

Its inverse is the transition function in the opposite direction:

$$\phi_{21}^{-1} = \phi_{12} = \varphi_1 \circ \varphi_2^{-1} : \varphi_2(U \cap V) \xrightarrow{\sim} \varphi_1(U \cap V)$$

Definition 2.4. Let X be a topological manifold. An **atlas** on X is a collection of co-ordinate charts on X :

$$\varphi_i : U_i \xrightarrow{\sim} \tilde{U}_i \subset \mathbb{R}^n$$

indexed by some, possibly infinite, set I , such that:

$$\bigcup_{i \in I} U_i = X$$

An atlas is a **smooth atlas** if and only if for any two co-ordinate charts in the atlas, the transition function

$$\phi_{ji} = \varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \xrightarrow{\sim} \varphi_j(U_i \cap U_j), \quad i, j \in I$$

is a smooth function, i.e. it is infinitely differentiable.

Definition 2.5. Let X be a topological manifold, and let

$$\mathcal{A} = \{(U_i, \varphi_i); i \in I\}$$

be a smooth atlas on X . Let (U, φ) be any coordinate chart on X . The chart (U, φ) is **compatible** with the atlas $\mathcal{A}$ if and only if the transition functions between (U, φ) and any chart in $\mathcal{A}$ are diffeomorphisms, i.e. they are smooth functions with smooth inverses.

In other words, the new chart (U, φ) is compatible with the atlas $\mathcal{A}$ if and only if the union $\mathcal{A} \cup \{(U, \varphi)\}$ is still a smooth atlas.

Let $\mathcal{A}, \mathcal{B}$ be smooth atlases on X . The atlases $\mathcal{A}$ and $\mathcal{B}$ are **compatible** if and only if every chart in $\mathcal{A}$ is compatible with $\mathcal{B}$, and vice versa.

Corollary 2.6 ([2, Corollary 2.20]). *Compatibility is an equivalence relation on smooth atlases.*

Definition 2.7. A **smooth manifold** is a topological manifold X together with an equivalence class $[\mathcal{A}]$ of compatible smooth atlases on X .

The equivalence class $[\mathcal{A}]$ of compatible smooth atlases is called a **smooth structure** on X .

Definition 2.8. Let X, Y be smooth manifolds, of dimension m and n respectively, and let

$$H : X \longrightarrow Y$$

be a continuous function. Fix a point $x \in X$. H is **smooth at x** if and only if there exists a chart (U, φ) compatible with the smooth structure on X , with $x \in U$, and a chart (V, ψ) compatible with the smooth structure on Y , with $H(U) \subset V$, such that the composition

$$\psi \circ H \circ \varphi^{-1} : \varphi(U) \longrightarrow \psi(V)$$

is smooth at $\varphi(x)$.

The function H is **smooth** if and only if it is smooth at every point $x \in X$.

Definition 2.9. Let X, Y be smooth manifolds. A function

$$H : X \longrightarrow Y$$

is a **diffeomorphism** if and only if H is a bijection, and both H and H^{-1} are smooth.

If there exists a diffeomorphism between X and Y , they are **diffeomorphic**.

Definition 2.10. Following Segal, let X be a set. A **pseudo-chart** on X is the pair (U, φ) , where $U \subset X$ and

$$\varphi : U \longrightarrow \varphi(U)$$

is a bijection onto an open subset $\varphi(U) \subset \mathbb{R}^n$.

A collection $\mathcal{A} = \{(U_i, \varphi_i) : i \in I\}$ of pseudo-charts, with the same dimension n , is called a pseudo-atlas on X if

$$\bigcup_{i \in I} U_i = X.$$

Proposition 2.11 ([2, Proposition 2.26]). *Let X be a set, and let $\mathcal{A} = \{(U_i, \varphi_i) : i \in I\}$ be a pseudo-atlas for X . Suppose that for any pair $(U_1, \varphi_1), (U_2, \varphi_2)$ of pseudo-charts in $\mathcal{A}$, we have that:*

1. *both $\varphi_1(U_1 \cap U_2) \subset \varphi_1(U_1)$ and $\varphi_2(U_1 \cap U_2) \subset \varphi_2(U_2)$ are open subsets, and*
2. *the transition function is continuous.*

Then there exists a unique topology on X such that each pseudo-chart (U_i, φ_i) is a co-ordinate chart on X .

Corollary 2.12 ([2, Corollary 2.28]). *If we further assume that each transition function ϕ_{21} is smooth, then $[\mathcal{A}]$ is a smooth structure on the topological manifold X , provided the resulting topology is Hausdorff and second-countable.*

Definition 2.13. Let X be a smooth manifold of dimension n , let m be an integer satisfying $0 \leq m \leq n$, and let $\mathcal{M} \subset X$. The subset $\mathcal{M}$ is an **m-dimensional submanifold** of X if and only if, for every $x \in \mathcal{M}$, there exists a compatible co-ordinate chart

$$\varphi : U \xrightarrow{\sim} \varphi(U) \subset \mathbb{R}^n$$

with $x \in U$, and an m -dimensional affine subspace $A \subset \mathbb{R}^n$, such that

$$\varphi(U \cap \mathcal{M}) = \varphi(U) \cap A.$$

This is the convention of [2, Definition 3.2]. We also use the term **embedded submanifold** for a submanifold in this sense.

By [2, Proposition 3.7], $\mathcal{M}$, with the subspace topology, carries a smooth structure induced from X . We always equip a submanifold with this structure.

Definition 2.14. Let $\mathcal{M}$ be a smooth manifold, and let $x \in \mathcal{M}$. A curve through x is a smooth function $c : I \rightarrow \mathcal{M}$, where I is an open interval containing 0 and $c(0) = x$. Two curves c_1, c_2 through x are **tangent at x** if and only if, for any compatible co-ordinate chart (U, φ) containing x ,

$$\left. \frac{d}{dt}(\varphi \circ c_1)(t) \right|_{t=0} = \left. \frac{d}{dt}(\varphi \circ c_2)(t) \right|_{t=0}.$$

The compositions are taken on sufficiently small intervals around 0. This condition is independent of the choice of chart.

A **tangent vector** at x is an equivalence class of curves through x under this relation. The **tangent space** at x is the set

$$T_x \mathcal{M} = \{\text{smooth curves through } x\} / \sim,$$

where $c_1 \sim c_2$ means that the curves are tangent at x , following [2, Definitions 5.4–5.5].

By [2, Proposition 5.8], $T_x \mathcal{M}$ is a vector space of dimension $\dim \mathcal{M}$. Its vector space structure is the one for which the chart identification

$$[c] \mapsto \left. \frac{d}{dt}(\varphi \circ c)(t) \right|_{t=0}$$

is a linear isomorphism with $\mathbb{R}^{\dim \mathcal{M}}$.

For an embedded submanifold $\mathcal{M}$ of a finite-dimensional real vector space $\mathcal{E}$, we identify $[c]$ with the ambient velocity $c'(0)$, as in [2, Section 5.2]. Under this identification, $T_x\mathcal{M}$ is a subspace of $\mathcal{E}$, and

$$T_x\mathcal{M} = \{c'(0) : c : I \longrightarrow \mathcal{M} \text{ is smooth and } c(0) = x\}.$$

Here, derivatives are taken in $\mathcal{E}$. If $\mathcal{M}$ is open in $\mathcal{E}$, then $T_x\mathcal{M} = \mathcal{E}$.

Definition 2.15. Let X, Y be smooth manifolds, and let $F : X \longrightarrow Y$ be smooth. The **derivative** of F at $x \in X$ is the linear map

$$DF(x) : T_xX \longrightarrow T_{F(x)}Y, \quad [c] \longmapsto [F \circ c].$$

This map is well-defined and linear by [2, Proposition 5.10].

Theorem 2.16 (Regular level sets [2, Proposition 3.16 and Lemma 5.13]). *Let $\mathcal{E}$ and $\mathcal{E}'$ be finite-dimensional real vector spaces, and let $F : \mathcal{E} \rightarrow \mathcal{E}'$ be smooth. A point $c \in \mathcal{E}'$ is called a regular value of F if $DF(x)$ is surjective for every $x \in F^{-1}(c)$.*

If c is a regular value and $\mathcal{M} = F^{-1}(c)$ is nonempty, then $\mathcal{M}$ is a smooth embedded submanifold of $\mathcal{E}$, with

$$\dim \mathcal{M} = \dim \mathcal{E} - \dim \mathcal{E}', \quad T_x\mathcal{M} = \ker DF(x) \quad \text{for every } x \in \mathcal{M}.$$

To define gradients on these manifolds, we equip their tangent spaces with inner products.

Definition 2.17. Following [1, Definitions 3.51–3.53], let $\mathcal{M}$ be a smooth manifold. A **Riemannian metric** on $\mathcal{M}$ assigns an inner product $\langle \cdot, \cdot \rangle_x$ to each tangent space $T_x\mathcal{M}$, varying smoothly with x . In co-ordinates (U, φ) , this means that each function

$$x \longmapsto \left\langle (D\varphi(x))^{-1}e_i, (D\varphi(x))^{-1}e_j \right\rangle_x$$

is smooth on U , where e_i, e_j are standard basis vectors of $\mathbb{R}^{\dim \mathcal{M}}$. A smooth manifold equipped with a Riemannian metric is a **Riemannian manifold**.

Definition 2.18. Let $\mathcal{M}$ be an embedded submanifold of a finite-dimensional real vector space $\mathcal{E}$ equipped with an inner product $\langle \cdot, \cdot \rangle_{\mathcal{E}}$. The induced metric on $\mathcal{M}$ is defined by

$$\langle u, v \rangle_x = \langle u, v \rangle_{\mathcal{E}}, \quad u, v \in T_x\mathcal{M}.$$

This is a Riemannian metric by [1, Proposition 3.54]. Equipped with this metric, $\mathcal{M}$ is a **Riemannian submanifold** of $\mathcal{E}$.

Definition 2.19. Let $\mathcal{M}$ be a Riemannian manifold, and let $f : \mathcal{M} \rightarrow \mathbb{R}$ be smooth. For each $x \in \mathcal{M}$, the **Riemannian gradient** of f at x is the unique vector $\text{grad } f(x) \in T_x\mathcal{M}$ satisfying

$$Df(x)[v] = \langle \text{grad } f(x), v \rangle_x \quad \text{for every } v \in T_x\mathcal{M}.$$

Here,

$$Df(x)[v] = \left. \frac{d}{dt} f(c(t)) \right|_{t=0},$$

where c is any smooth curve through x representing v . This value is independent of the choice of such a curve.

Definition 2.20. Let $\mathcal{M}$ be an embedded submanifold of a finite-dimensional real vector space $\mathcal{E}$ equipped with an inner product $\langle \cdot, \cdot \rangle$, and let $x \in \mathcal{M}$. The **orthogonal projector** to $T_x\mathcal{M}$ is the linear map $\text{Proj}_x : \mathcal{E} \rightarrow \mathcal{E}$ characterized by the following properties:

1. Range: $\text{im}(\text{Proj}_x) = T_x\mathcal{M}$.
2. Projector: $\text{Proj}_x \circ \text{Proj}_x = \text{Proj}_x$.
3. Orthogonal: $\langle u - \text{Proj}_x(u), v \rangle = 0$ for every $v \in T_x\mathcal{M}$ and $u \in \mathcal{E}$.

For an open submanifold, Proj_x is the identity because $T_x\mathcal{M} = \mathcal{E}$.

Proposition 2.21 ([1, Proposition 3.61]). *Let $\mathcal{M}$ be a Riemannian submanifold of a finite-dimensional real vector space $\mathcal{E}$ equipped with an inner product $\langle \cdot, \cdot \rangle$, and let $f : \mathcal{M} \rightarrow \mathbb{R}$ be smooth. Then the Riemannian gradient of f is given by*

$$\text{grad } f(x) = \text{Proj}_x(\nabla \bar{f}(x)), \quad x \in \mathcal{M},$$

where $\bar{f}$ is any smooth extension of f to an open neighbourhood of $\mathcal{M}$ in $\mathcal{E}$, and $\nabla \bar{f}(x)$ is its Euclidean gradient.

The Euclidean gradient is characterized by

$$D\bar{f}(x)[v] = \langle \nabla \bar{f}(x), v \rangle \quad \text{for every } v \in \mathcal{E}.$$

A gradient gives a tangent direction at a point. Retractions allow us to use this direction to move on the manifold.

Definition 2.22. Let $\mathcal{M}$ be a smooth manifold. The **tangent bundle** of $\mathcal{M}$ is the set

$$T\mathcal{M} = \{(x, v) : x \in \mathcal{M}, v \in T_x\mathcal{M}\}.$$

It carries a natural smooth structure by [2, Proposition E.1]. A **retraction** on $\mathcal{M}$ is a smooth map

$$R : T\mathcal{M} \rightarrow \mathcal{M} : (x, v) \mapsto R_x(v)$$

such that, for every $x \in \mathcal{M}$ and $v \in T_x\mathcal{M}$, the curve $c(t) = R_x(tv)$ satisfies $c(0) = x$ and represents v .

For an embedded submanifold $\mathcal{M} \subset \mathcal{E}$, we regard R as a map into $\mathcal{E}$. It is smooth if it extends to a smooth map on an open neighbourhood of $T\mathcal{M}$ in $\mathcal{E} \times \mathcal{E}$.

A manifold may admit more than one retraction.

Intuitively, a retraction provides a way to move from a point on the manifold along a tangent vector while staying on the manifold.

Definition 2.23. Let X, Y be smooth manifolds. A smooth function $F : X \rightarrow Y$ is a **submersion** if and only if

$$\text{rank } DF(x) = \dim Y \quad \text{for every } x \in X.$$

Equivalently, $DF(x) : T_xX \rightarrow T_{F(x)}Y$ is surjective at every point, as in [2, Definition 4.16 and Section 5.1].

2.2 Examples

Example 2.24.

Example 2.25.

3 Motivation

4 The Stiefel Manifold

Definition 4.1.

Remark 4.2.

5 The Grassmann Manifold

Lemma 5.1.

Proof.

□

Proposition 5.2.

Proof.

□

6 From Stiefel to Grassmann

Theorem 6.1.

Proof.

□

Corollary 6.2.

Proof.

□

7 Applications

7.1 $\text{Gr}(2, 4)$ in full

7.2 PCA on the Grassmannian

8 Discussion

9 Further Directions

10 Conclusion

References

- [1] Nicolas Boumal, *An introduction to optimization on smooth manifolds*, Cambridge University Press, 2023.
- [2] Ed Segal, *Manifolds*, 2016, Lecture notes, University College London.